



Introduction

PB177 Cyber Attacks

September 17, 2026

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Course organization

Instructors – CYBERSEC laboratory

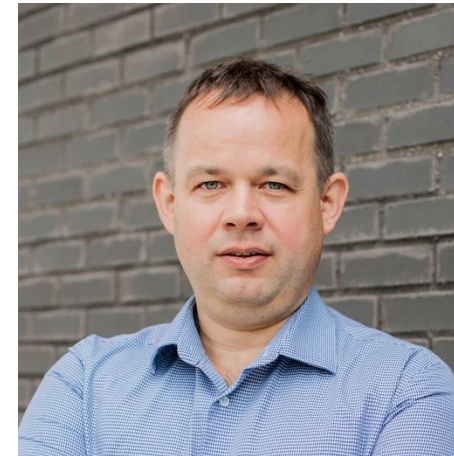
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founder of the incident response
team at MU (CSIRT-MU) in 2009,
KYPO, CYBERSEC, INJECT



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15+ years of cybersecurity experience
CSIRT-MU, Flowmon, KYPO,
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Goal of this course

- To introduce the **lifecycle of cyber attacks** and typical **attack actions within the corporate network environment**, including their taxonomies.
- To **practice executing attacks** and explore options for **their analysis and defense** in a secure environment (sandbox).
- A core course of the [Cybersecurity](https://cybersec.fi.muni.cz) bachelor's degree programme.

Course prerequisites

IS MU: PB156 **Computer Networks** && PB156cv **Computer Networks – Practicals** && PV004 **UNIX** && ! PV276 Seminar on Cyber Attacks

In particular:

- the TCP/IP stack, network communication using common protocols such as IP, TCP, UDP, HTTP, DNS, SSH
- Linux command line: navigating file system, running application and services
- Git(lab) essentials
- Careful reading of instructions

Previous experience with Docker/Podman is beneficial, but not necessary.

Target audience of this course

- Students of Cybersecurity bachelor's degree programme
 - Recommended to study in the 3rd semester, along with PV210 Cybersecurity in an Organization
- Students who would like to learn common attackers' techniques
- Students with **elementary** knowledge and skills from IT (cyber) security, development and operations

Topics NOT covered in this course

- Attacks in the context of an organization, cybersecurity incident handling. Addressed in PV210.
- Penetration testing **process and practice**. Part of a PA211.

Course format

- Lectures (1 hour).
 - Lab sessions (seminars, 2 hours, once per 2 weeks).
 - Homework assignments.
 - Exam.
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- This is a hands-on course. Less theory, more practice.

Course schedule

Week	Date	Topic	Seminar (lab-groups)	Assigned	Deadline
1	17/09	Introduction	1-01,02,03		
2	24/09	Attack lifecycle, MITRE ATT&CK, Reconnaissance	1-04,05,06		
3	01/10	MITRE ATT&CK - Resource Development, Initial Access	2-01,02,03		
4	08/10	MITRE ATT&CK - Execution, Persistence, Privilege Escalation	2-04,05,06	HW1	
5	15/10	<i>no lecture and seminars this week</i>	---		
6	22/10	MITRE ATT&CK - Defense Evasion, Credential Access	3-01,02,03		
7	29/10	MITRE ATT&CK - Discovery, Lateral Movement, Collection	3-04,05,06	HW2	HW1
8	05/11	MITRE ATT&CK - C&C, Exfiltration, Impact	4-01,02,03		
9	12/11	Advanced Persistent Threat	4-04,05,06	HW3	HW2
10	19/11	Web Application Attacks	5-01,02,03		
11	26/11	MITRE D3FEND I	5-04,05,06	HW4	HW3
12	03/12	MITRE D3FEND II, MITRE Engage	6-01,02,03		
13	10/12	Penetration Testing	6-04,05,06		
14	17/12	Feedback Session	7-01-06		HW4

Note irregularity in even/odd weeks from Lab 3 on.

Lectures

- Introduces concepts, techniques, tools.
- Presents existing attacks and campaigns.
- You may learn all these knowledge from online sources, however it is easier to attend the lecture.
- If you have any questions during lectures, feel free to ask during lectures or use an online tool (see further).

Real-time questions during class – Claper

- You can use Claper for asking questions during the lecture.
- We will answer your questions at the end of the lecture.
- Go to:
<https://claper.fi.muni.cz/e/pb177>



Lab sessions (seminars) – I

- Starts with **KvIS** refreshing terms from lectures.
- Continues with a **summary of key knowledge** for the lab session.
- Each of you will work in your **own sandbox** (safe virtual environment).
- Sandbox configuration is provided in a **Gitlab repo**.
 - If you haven't received an invitation from Gitlab yet, email vykopal@fi.muni.cz immediately.
- You will complete several tasks (= lab) at your own pace.
- Task assignments and solutions are presented at **INJECT Exercise Platform**.
 - If you haven't received credentials for the platform, email vykopal@fi.muni.cz immediately.
- If you have any questions during the seminar, ask an instructor in person.

Lab sessions (seminars) – II

- Sandboxes were prepared and **tested at nymfe hosts at FI.**
- Sandboxes consists of **containers orchestrated by Podman.**
- This is the **only HW and SW configuration we guarantee** the sandbox works.
- If you prefer using your own laptop, install Podman or Docker and start the sandbox according to [instructions in Gitlab](#). We will **NOT troubleshoot your setup during seminars.**
- We have successfully run the sandboxes with Podman and Docker on Windows and macOS too. Docker is recommended for Windows and macOS.

Attacks in this course and outside

- In this course, you will practice various attacks within a controlled sandbox environment.
- You are fully responsible for any actions taken outside of this designated sandbox environment.
- **Unauthorized activities conducted outside of this environment are strictly prohibited and might have legal consequences.**

Czech Criminal Code

- § 230 Neoprávněný přístup k počítačovému systému a nosiči informací
- § 231 Opatření a přechovávání přístupového zařízení a hesla k počítačovému systému a jiných takových dat
- § 232 Poškození záznamu v počítačovém systému a na nosiči informací a zásah do vybavení počítače z nedbalosti
- See ["Trestní zákoník"](#).

Homework assignments

- Requires skills exercised in seminars.
- 4 homeworks (HW)
- 2 weeks to complete HW1, 2, and 3
- 3 weeks for HW4
- Individual assignments. Teamwork is forbidden.
- Submit answers resulting from your own work.
- Prefer using our CyberBrain assistant to others (ChatGPT etc.)

CyberBrain – CYBERSEC AI assistant

- In-house AI assistant, interface to locally hosted models: GPT OSS, GLM, Gemma4, and DeepSeek v4 models enriched by course materials (e.g., slides).
- Available at <https://cyberbrain.fi.muni.cz>. Log in using Unified login MUNI and your UČO and primary password.
- Feel free to use it when solving labs or homeworks.
- **You will get limited access to CyberBrain at the exam.**

Exam

- **Two parts:** key terms and concepts (as KvIS) + hands-on part.
- The hands-on part assesses whether you can use tools for completing **practical tasks** practised in seminars and homeworks. Held in an **exam sandbox**.
- **Closed-book exam**, except for a limited use of the CyberBrain AI assistant and a cheatsheet for the hands-on part.
- Other AI assistants, online resources, and real-time communication with human beings are forbidden.
- The best preparation is the completion of all seminars and homeworks.
- No early exam date, the first exam date will be in January.

Why closed-book exam and limited AI?

- We believe that **relying on AI leads to learning less** than one could otherwise.
- Sometimes you can also get **inefficient or misleading answers**.
- The exam format pushes you to **understand** the concepts and **have some practical experience** before the exam.
- Still, you can use an AI assistant, but wisely.
- Evidence: [The Generative AI Learning Penalty: Evidence from Chinese Secondary Education](#): AI adoption raises homework scores and reduces completion time, but lowers exam scores.

Grading

A-E, F grades based on points (pts):

- **Lab completion** (12%) – 2 points for each lab, 12 pts in total.
- **Homework** assignments (36%) – $4 \times 9 = 36$ pts.
- Only students who gained **24 pts from labs and homeworks can enrol to the exam.**
- **Exam** (52%)
 - Key terms and concepts – 16 points, **min. 4 pts** to pass the course.
 - Hands-on part – 36 points, **min. 18 pts** to pass the course.
- **Bonuses** (1–3 points above 100) – see next slide

Grading – Bonuses

- **Top-performing students in all KvISes** – up to 3 points.
 - Students with a top 9 total KvIS [score](#) (not KVIS points) will be awarded 3, 3, 3, 2, 2, 2, 1, 1, and 1 points.
- **Active participation in a feedback session** at the end of the semester – 1 point
 - Come and share your feedback with us.
- Your **grade can be improved** by bonus points only **if you have already passed 50-point threshold**. E. g. if you have 49 points and 3 bonus points, you will get F. If 89 and 1 bonus, then A.

Grading – Scale

Grading scale:

- **A** – 90 and more,
- **B** – 80–89,
- **C** – 70–79,
- **D** – 60–69,
- **E** – 50–59 (note thresholds from individual parts),
- **F** – 49 and less.

Examples of points and F grades:

- 23 pts from labs and HW, no exam
- 24 pts from labs and HW, 17 pts from hands-on part of the exam
- 3 pts from knowledge part of the exam

Grading – Your involvement – I

Getting the grade you want:

A

- Follow the rules, comply with instructions on the homework and exams.
- **Attend the seminars and complete labs** (at home).
- **Ask questions at seminars** when you're stuck.
- Read the instructions on the homework **very carefully**.
- Start **working on homework early**, not last night before the deadline.
- Submit **all homeworks**. Use CyberBrain if needed.
- **Revisit labs and homeworks** before the **exam**.

Grading – Your involvement – II

B–E

- Do not ask at seminars. Do not attend the seminars nor complete the labs.
- Submit 3 or 2 homeworks. Do not read the assignment instructions carefully.
Use AI assistants to get the solution. Do not try to understand the concepts and tools.
- **Relying on the AI assistant** during the exam.

F

- **Do only the minimum amount of work you think you need to pass.**
- Submit 1 or 2 homeworks.
- **Break the rules** when submitting homeworks or taking exam.

Any questions on course organization?

Key terms

Cyber attack

- [NIST](#) ([one of a few](#)): an attempt to gain **unauthorized** access to system services, resources, or information, or an attempt to **compromise** system integrity, availability, or confidentiality.
- [Australian Cyber Security Centre](#): a **deliberate** act through cyberspace to manipulate, disrupt, deny, degrade or destroy computers or networks, or the information resident on them, with the effect of seriously compromising national security, stability or economic prosperity.
- [ENISA](#): a cyber incident triggered by **malicious intent** where damages, disruptions or dysfunctionalities are caused.

Examples of cyber attacks

Have you heard about any cyber attacks?

Examples of common cyber attacks

- **Phishing** – stealing important information such as credit card numbers and passwords through e-mails involving social engineering and deception.
- **Ransomware** – taking control of a target's assets and demanding a ransom in exchange for the return of the asset's availability.
- **Distributed Denial of Service (DDoS)** – targeting system and data availability by exhausting the service and its resources or by overloading the network infrastructure.
- Source: [ENISA Threat Landscape 2025](#)

Is this a cyber attack?

1. IS MU not responding at 5.00pm because a lot of students try to enrol to seminar groups.
2. Paralyzed airports due to an error in CrowdStrike Falcon update resulting in crash and blue screen of death of Microsoft Windows machines. More details: [CrowdStrike Blog](#).

Answers

1. No.

2. No.

In both cases, there was **no malicious intent** or attempt to compromise systems.

Both can be considered **incidents**.

Cybersecurity incident

- NIST: A cybersecurity event that has been determined to **have an impact** on the organization prompting **the need for response and recovery**.
- Australian Cyber Security Centre: An unwanted or unexpected cyber security event, or a series of such events, that has either **compromised business operations or has a significant probability** of compromising business operations.

Cybersecurity incident

- ENISA: Any occurrence **that has impact on** any of the components of the cyber space or on the functioning of the **cyber space**, independent if it's natural or human made; **malicious or non-malicious intent**; deliberate, accidental or due to incompetence; due to development or due to operational interactions.

Cybersecurity event

- [NIST](#): A cybersecurity change that **may** have an impact on organizational operations (including mission, capabilities, or reputation).
- [Australian Cyber Security Centre](#): An occurrence of a system, service or network state **indicating a possible** breach of security policy, failure of safeguards or a previously unknown situation that may be relevant to security.
- Examples: detection of an unknown device connected to the network, multiple failed login attempts, unplanned service outage

Defender's view

- **Events** are reported by third-party and internal detection systems, or users.
- Events may indicate **incidents** if business operations are compromised.
- If incidents are result of a **malicious intent**, they are **attacks** caused by **attackers** (adversaries).

Cyber threat

- NIST: The potential source of an adverse event.
- Australian Cyber Security Centre: Any circumstance or event with the potential to harm systems, applications or data.
- The terms **threat** and **attack** are often used interchangeably.
- What is the **difference** between **threat** and **attack**?

Cyber threat vs. attack

- NIST: The **potential** source of an adverse event.
- Australian Cyber Security Centre: Any circumstance or event with the **potential** to harm systems, applications or data.
- **Attack** has already been **carried out** or **is in progress**.
- **Threat** is any possible danger **that could result in an attack**.
- Example: phishing in general is a threat, but concrete instance of phishing affecting a particular organization/service and its users is an attack.

Vulnerability

- Sources of cyber threats.
- A software **vulnerability** is:
 - an **instance of** one or more **weaknesses** in a product that can be **exploited**, causing a **negative impact** to confidentiality, integrity, or availability; a set of conditions or behaviors that allows the violation of an explicit or implicit security policy. [[CVE.org](https://cve.org)]
 - a mistake in software that can be **directly** used by a hacker to gain access to a system or network. [cwe.mitre.org]
- Example: PHP Remote Code Execution ([CVE-2024-4577](https://cve.org/CVE-2024-4577)) – bypassing the escape process and injecting a command

Weakness

- A **weakness** is a condition in a software, firmware, hardware, or service component that, under certain circumstances, **could contribute to the introduction of vulnerabilities**. [cwe.mitre.org]
- Example: Improper Neutralization of Special Elements used in an OS Command ('OS Command Injection', [CWE-78](#)) is an essence of CVE-2024-4577.

From weakness to attack – I

- Software contains unique **vulnerabilities**, instances of **weaknesses**.
- Attacker **exploits** the concrete **vulnerability** to conduct concrete **attack**.
- Vulnerabilities enable **threats** to be successful attacks.
- There are many **threats** endangering organizations (defenders), threats can result in an **attack**.

From weakness to attack – II



Source: https://cwe.mitre.org/images/example_product_lifecycle.png

Common Vulnerabilities and Exposures (CVE)

- CVE are **unique, common identifiers** and records for **publicly known vulnerabilities**.
- CVE are catalogized and freely available at [CVE.org](https://cve.org).
- Each CVE record includes:
 - **CVE ID** number consisting of a year and four or more sequence digits, e.g., [CVE-2021-44228](#) or [CVE-2024-4577](#)
 - a brief **description** of the security vulnerability or exposure,
 - any pertinent **references**, i.e., vulnerability reports and advisories).
- **Question for you:** how many CVEs were published in 2025?

Common Vulnerabilities and Exposures (CVE)

48,185 CVE records published in 2025.

Common Vulnerability Scoring System (CVSS)

- An open framework for communicating the **characteristics and severity of vulnerabilities** developed by FIRST.
- Its output is **numerical scores from 0.0 to 10.0** indicating the severity of a vulnerability relative to other vulnerabilities.
- The latest [CVSS version is 4.0](#), commonly used is version 3.1.
- Example: CVSS 3.1 of [CVE-2024-4577](#) is 9.8 Critical, 4.0 not provided.

National Vulnerability Database

- The U.S. government repository providing **comprehensive information and tools for managing vulnerabilities**.
- Includes databases of security checklist references, security-related software flaws, product names, and impact metrics.
- Uses **CVE and CWE IDs**.
- <https://nvd.nist.gov/vuln/search>

Terms and abbreviations covered today

- Event
- Incident
- Attack
- Weakness
- Vulnerability
- Threat
- CVE
- CVSS
- NVD

Listening

- **Darknet Diaries** – podcast about hackers, breaches, shadow government activity, hacktivism, and cybercrime:

<https://darknetdiaries.com>



Other announcements – Kybersoutěž

- A new year of Czech national cybersecurity competition has been started.
- Open for primary, high school and **university students** aged 9–**24**.
- Register at <https://www.kybersoutez.cz> now.



Other announcements – MUNI CTF Meetup

Hack with [CSIRT-MU](#) and your peers at monthly meetups.

When: Friday, September 18, 12:00–20:00

Where: Meeting Room 15.21 in **Šumavská Tower A** (15th floor)

What do you need: Kali Linux VM and a free account on [HackTheBox](#)

For more information how to get there contact Jiří Rája at raja@ics.muni.cz.

Questions, answers, and feedback in Claper

- You can use Claper for asking questions during the lecture.
- We will answer your questions at the end of the lecture.
- Go to:
<https://claper.fi.muni.cz/e/pb177>
- **Please provide us with feedback:**
How can today's class be improved?



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